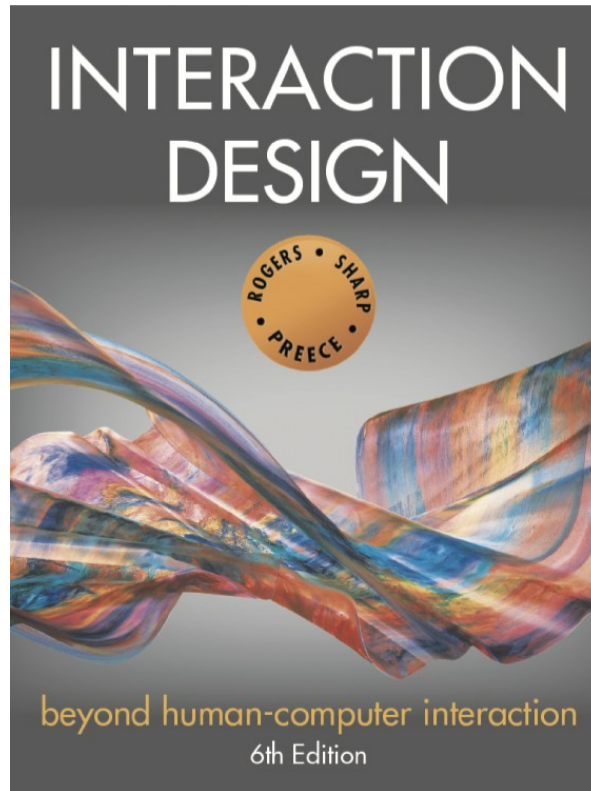




Chapter 1

WHAT IS INTERACTION DESIGN?

Learning Objectives

- Explain the difference between **good and poor interaction design**.
- Consider the pros and cons of **transforming activities to become digital**.
- Describe what **interaction design** is and how it relates to **human-computer interaction** and other fields.
- Explain the relationship between the **user experience and usability**.
- Introduce what is meant by **accessibility** and **inclusiveness** in relation to human-computer interaction.

Learning Objectives (cont.)

- Describe **what and who is involved** in the process of interaction design.
- Outline the different **forms of guidance** used in interaction design.
- Enable you to **evaluate an interactive product** and explain what is good and bad about it in terms of the goals and core principles of interaction design.

What's in a Name?

Interaction Design can have different perspectives

- User-centered design
 - Human-centered design
 - People-centered design
 - Customer experience design
-
- Which do you think is preferable and why?

People-Centered-Design (Don Norman)

- People-centered design is a practice where designers focus on specific people's needs, taking the time to learn from particular populations.
- “**People-centered**” is a **human-centered** design principle.

“When you design, you have to understand what the capabilities are of the people you're designing for.”

Don Norman

<https://www.interaction-design.org/literature/topics/people-centered-design>

4 Principles of Human-Centered Design



People-Centered



**Solve the
Right Problem**



**Everything is
a System**



**Small & Simple
Interventions**

4 Components of Human-Centered Design

1. Ensuring that we solve the core, root issues, not just the problem as presented to us (which is often the symptom, not the cause)
2. Focusing on people
3. Taking a systems point of view, realizing that most complications result from the interdependencies of the multiple parts
4. Continually testing and refining our proposals, ensuring they truly meet the needs of the people for whom they are intended

Good and bad design

TiVo is a **digital video recorder** that was originally developed to enable the viewer to record TV shows.

Why is the **TiVo remote control** much better designed than standard remote controls?

- Designed with the viewer in mind
- Peanut shaped to fit in hand
- Logical layout and color-coded, distinctive buttons
- Easy-to-locate buttons

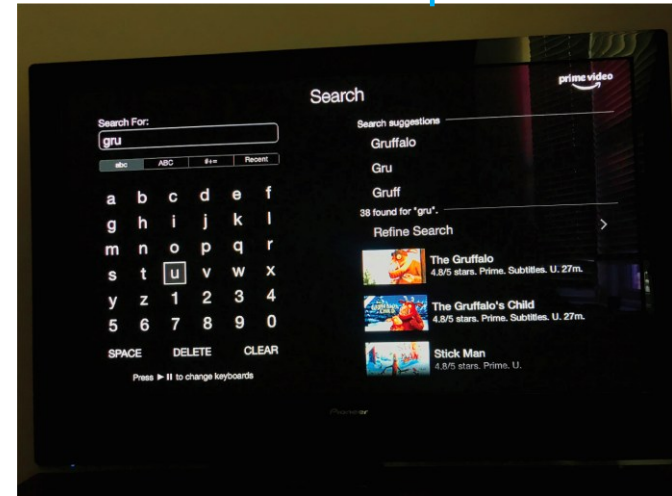


Dilemma

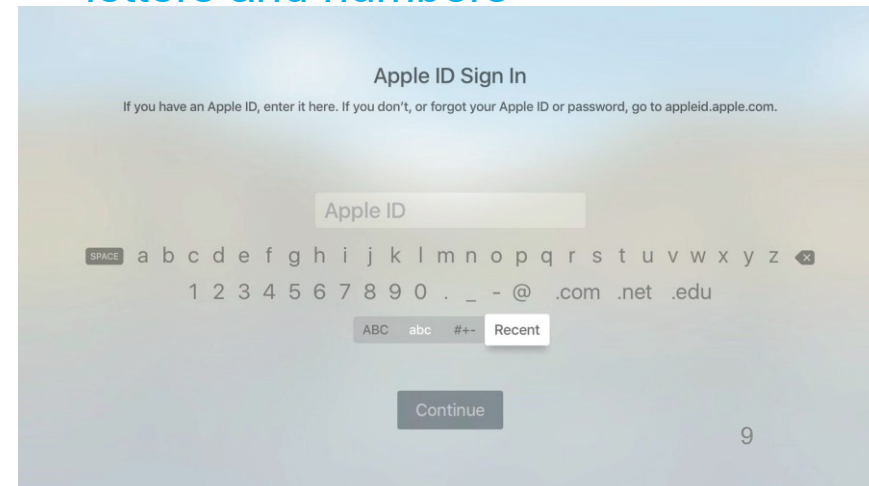
Which of the following is **the best way to interact with a smart TV**? Why?

1. Pecking using **a grid keyboard** via a remote control
2. Swiping across **two alphanumeric rows** using a touchpad on a remote control (e.g., Apple TV)
3. **Voice control** using remote or smart speaker, e.g., amazon echo

(a) by selecting letters and numbers from a square matrix



(b) by swiping along a single line of letters and numbers



Switching to Digital

- Many activities that used to be done with a physical artifact have gone digital
 - e.g. ticket machines, parking meters, ATMs
 - Nowadays, we can use Apps, CCs, QR codes, etc.)
- Benefits
 - Makes many tasks easier, quicker and more convenient
 - Our details can also be stored ready for next time
 - Don't have to wait in line before when buying a physical ticket
 - Easy to swipe QR code to gain entrance through a ticket barrier
- Disadvantages
 - Not everyone has a modern smartphone
 - Some people still prefer to use older phones which the apps won't work on
 - Some people prefer not to divulge personal information online
 - Some people prefer to talk to real people when making a purchase

Switching to Digital

Figure 1.3 The form used for a parking app in the United Kingdom. It takes five seconds to complete and can be done while sitting in the car.

The screenshot shows a mobile app interface for confirming parking. At the top, the status bar displays the time 13:19 and signal/battery icons. The app header is 'Confirm Parking' with a back arrow. The main content area is divided into several sections: 1. Location: A box containing a location pin icon, the text '805805 Stanmer Park - Outer Car Parks Brighton & Hove City Council - Off Street', and a green checkmark icon. 2. License Plate: A box containing a car icon, the text 'GY14FG', and a dropdown arrow icon. 3. Duration: A box containing a clock icon, the text 'Expires today at 14:19', and 'Parking for 1 hour', with a green checkmark icon. 4. Cost: A box containing a tag icon, the text 'Total cost £1.20', and 'Includes additional fees (+£0.20)'. Below these is a decorative illustration of a city street with buildings and a tree. 5. Payment: A section titled 'Paying with' containing a box with 'Card ending *228' and a Mastercard logo. At the bottom is a large green button labeled 'PAY & PARK'.

13:19

Confirm Parking

805805
Stanmer Park - Outer Car Parks
Brighton & Hove City Council - Off Street

GY14FG

Expires today at
14:19
Parking for 1 hour

Total cost
£1.20
Includes additional fees (+£0.20)

Paying with

Card ending *228

PAY & PARK

What to design

Need to take into account:

- Who the users are
- What activities are being carried out
- Where interaction is taking place

Need to optimize the interactions people have with a product:

- So that they match their activities and needs
- Many interaction devices, many ways to interact

What to design

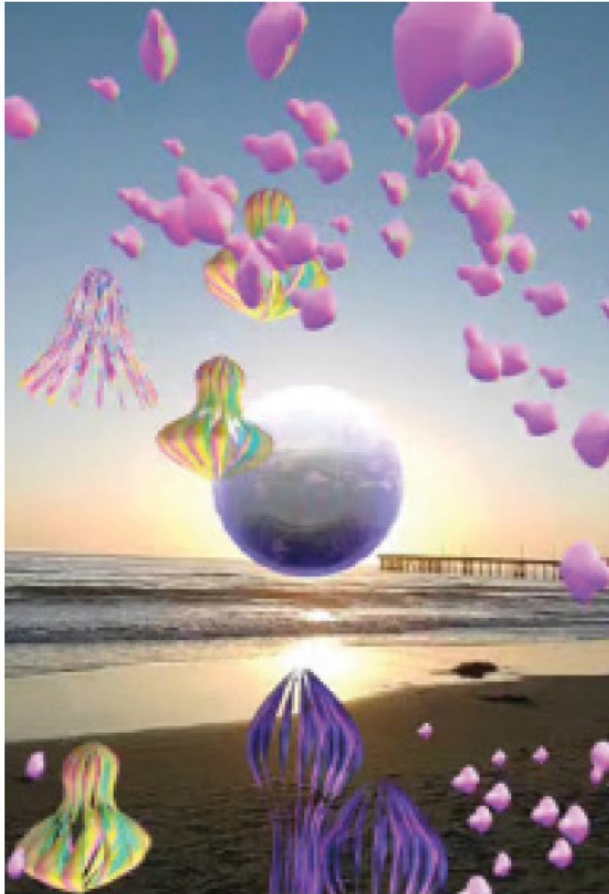


Figure 1.4 The digital world overlaying the physical experienced when wearing Snap AR Spectacles

What to design

- The advent of the **Internet of Things (IoT)**, where data is collected from sensors and travels via the Internet to other devices, has been embedded into several of our household products.
 - **For example**, a popular household IoT-enabled product is **home security**, where people can keep an eye on their home from the data relayed to their smartphone via a combination of sensors placed in their home. These include **motion detectors**, **glass breaking detectors**, and **smart object detectors**. A **video camera** can be attached to someone's doorbell and relayed to a smartphone app so the owner can check up on who has rung it—even though they may be on vacation.

What is interaction design?

By Interaction Design, we mean creating user experiences that enhance and augment the way people work, communicate and interact.

“Designing interactive products to support the way people communicate and interact in their everyday and working lives.”

Sharp, Rogers, and Preece (2019)

“The design of spaces for human communication and interaction.”

Winograd (1997)

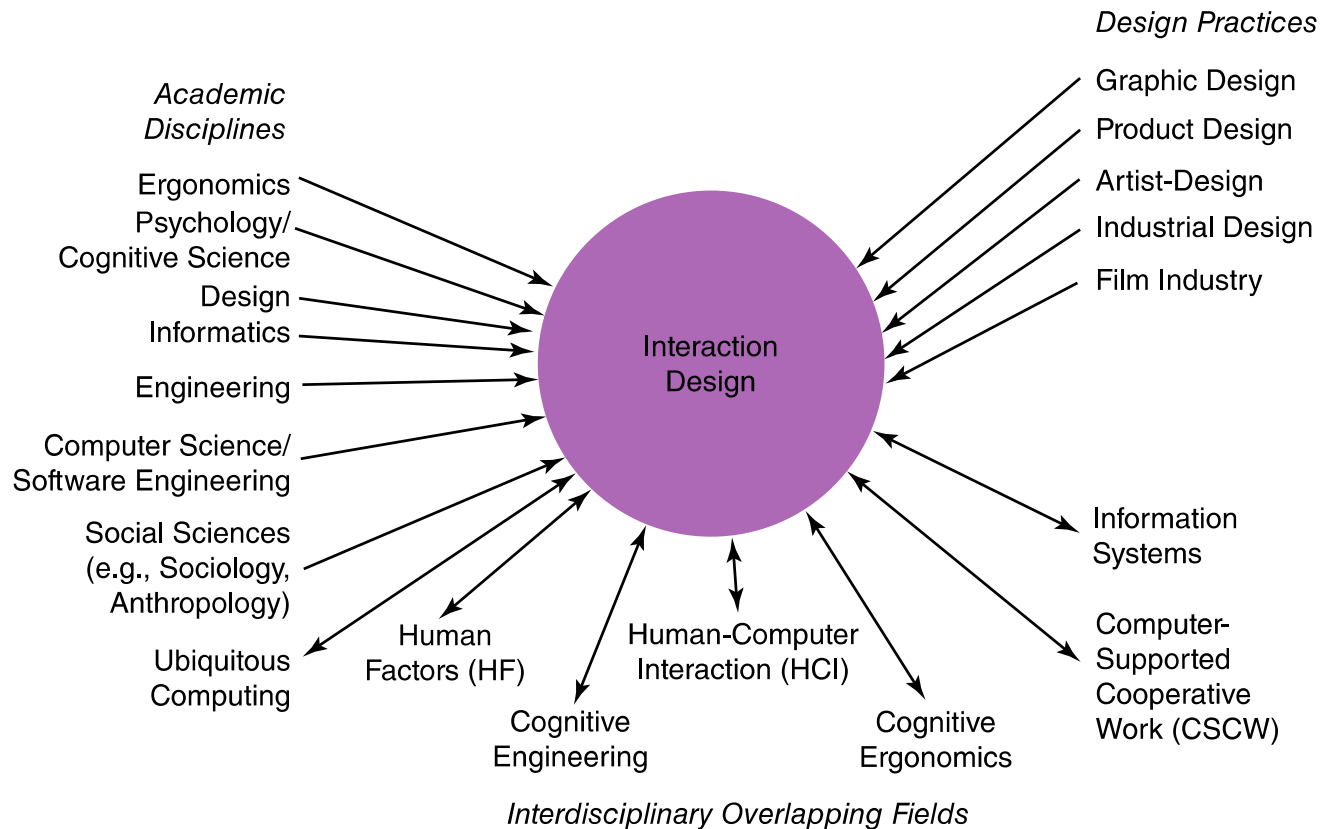
Goals of interaction design

- Develop usable products
- Involve people in the design process
- Consider what people are good and bad at
- Consider what might help people with the way they currently do things
- Think through what might provide quality experiences
- Consider a person's privacy concerns if data is being collected about them
- Listen to what people want and getting them involved in the design
- Use people-centered techniques during the design process

Which kind of design?

- Number of other terms used emphasizing what is being designed, for example:
 - User interface design, software design, user-centered design, product design, web design, experience design (UX)
- Interaction design is the umbrella term covering all of these aspects:
 - Fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

Interaction design (Fig. 1.5)



Relationship between ID, HCI, and other fields—academic disciplines

Academic disciplines contributing to ID:

- Psychology
- Social Sciences
- Computing Sciences
- Engineering
- Ergonomics
- Informatics

Relationship between ID, HCI and other fields—design practices

Design practices contributing to ID:

- Graphic design
- Product design
- Artist-design
- Industrial design
- Film industry

Relationship between ID, HCI and other fields—interdisciplinary fields

Interdisciplinary fields that 'do' interaction design:

- HCI
- Ubiquitous Computing
- Human Factors
- Cognitive Engineering
- Cognitive Ergonomics
- Computer Supported Co-operative Work
- Information Systems

Working in multidisciplinary teams

- As shown in Fig. 1.5, many people from different backgrounds involved in IxD
- Different perspectives and ways of seeing and talking about things
- Benefits
 - More ideas and designs generated
- Disadvantages
 - Difficult to communicate and progress forward the designs being create
 - Terms have different meaning in different disciplines

Interaction design in business

Interaction design has become widespread.

Large number of ID consultancies. Examples of well-known ones include:

- **Nielsen Norman Group:** “help companies enter the age of the consumer, designing human-centered products and services”
- **Cooper:** “From research and product to goal-related design”
- **IDEO:** “creates products, services and environments for companies pioneering new ways to provide value to their customers”

People-centered design

- involves understanding how people feel about a product and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it.
- their overall impression of how good it is to use
- the quality of the experience
- “It is not enough that we build products that function, that are understandable and usable, we also need to build joy and excitement, pleasure and fun, and yes, beauty to people's lives.”
 - Don Norman (2004)

Defining the user experience

How users perceive a product, such as whether a smartwatch is seen as sleek or chunky, and their emotional reaction to it, such as whether people have a positive experience when using it.

(Hornbæk and Hertzum, 2017)

Hassenzahl's (2010) model of the user experience

- Pragmatic: how simple, practical, and obvious it is for the user to achieve their goals
- Hedonic: how evocative and stimulating the interaction is to users
- Hassenzahl et al (2021) reflection on the way the user experience has evolved over the last 20 years
 - growing interest in designing for hedonic aspects in relation to wellbeing

Why was the iPod user experience such a success?



Figure 1.6 The iPod Nano Touch

Source: ©Press Association, reproduced with permission.

- Quality user experience from the start
- Simple, elegant, distinct brand, pleasurable, must have fashion item, catchy names, cool...

Core characteristics of interaction design

- Users should be involved throughout the development of the project
- Specific usability and user experience goals need to be identified, clearly documented, and agreed to at the beginning of the project
- Iteration is needed through the core activities

Why?

Help designers:

- Understand how to design interactive products that fit with what people want, need, and may desire
- Appreciate that one size does not fit all (for example, teenagers are very different to grown-ups)
- Identify any incorrect assumptions they may have about particular user groups. (for example, not all old people want or need big fonts)
- Be aware of both people's sensitivities and their capabilities

Accessibility and inclusiveness

Accessibility: the extent to which an interactive product is accessible by as many people as possible

- Focus is on people with disabilities; for instance, those using android OS or apple voiceover

Inclusiveness: making products and services that accommodate the widest possible number of people

- For example, smartphones designed for all and made available to everyone regardless of their disability, education, age, or income

Disabilities

- Whether someone is disabled changes over time with age, or recovery from an accident
- The severity and impact of an impairment can vary over the course of a day or in different environmental conditions
- Disabilities can result because technologies are designed to necessitate a certain type of interaction that is impossible for someone with an impairment

Understanding disability

Disabilities can be classified as:

- Sensory impairment (such as loss of vision or hearing)
- Physical impairment (having loss of functions to one or more parts of the body after a stroke or spinal cord injury)
- Cognitive (including learning impairment or loss of memory/cognitive function due to old age)

Each type can be further defined in terms of capability:

- For example, someone might have only peripheral vision, be color blind, or have no light perception

Impairment can be categorized:

- Permanent (for instance, long-term wheelchair user)
- Temporary (that is, after an accident or illness)
- Situational (for example, a noisy environment means that a person can't hear)

Being cool about disability

- Prosthetics can be designed to move beyond being functional (and often ugly) to being desirable and fashionable
- People now refer to “wearing their wheels,” rather than “using a wheelchair”



Fashionable leg cover designed by Alleles Design Studio

Cultural differences

5/21/20323 versus 21/5/2023?

- Which should be used for international services and online forms?
- Why is it that certain products, like smartphones, are universally accepted by people from all parts of the world, whereas people from different cultures react to websites differently?

Usability goals

- Effective to use (effectiveness)
- Efficient to use (efficiency)
- Safe to use (safety)
- Having good utility (utility)
- Easy to learn (learnability)
- Easy to remember how to use (memorability)
- Enjoyable to use (satisfaction)

Usability goals

- **Usability goals** are typically stated as **questions**
- Simply asking “Is the system easy to learn?” is not going to be very helpful.
- Asking about the usability of a product in a more detailed way—for example,
 - “How long will it take someone to figure out how to use the most basic functions for a new smartwatch;
 - how much can they capitalize on from their prior experience; and
 - how long would it take them to learn the whole set of functions?”—will elicit far more useful information.

Usability goals: Effectiveness

- **Effectiveness** refers to how good a product is at doing what it is supposed to do.
- **Question:** Is the product capable of allowing people to carry out their work successfully, access the information that they need, or buy the goods that they want?

Usability goals: Efficiency

- **Efficiency** refers to the way a product supports people in carrying out their tasks.
- **Question:**
 - How many steps does it take to complete a task?
 - How does storing a person's personal details make it more efficient?

Usability goals: Safety

- **Safety** involves protecting a person from dangerous conditions and undesirable situations.
- Making interactive products safer in this sense involves
 1. Preventing the user from making serious errors by reducing the risk of wrong keys/buttons being mistakenly activated
 2. providing people with various means of recovery should they make errors, such as an undo function
- **Question:**
 - What is the range of errors that are possible using the product, and
 - what measures are there to permit someone to recover easily from them?

Usability goals: Utility

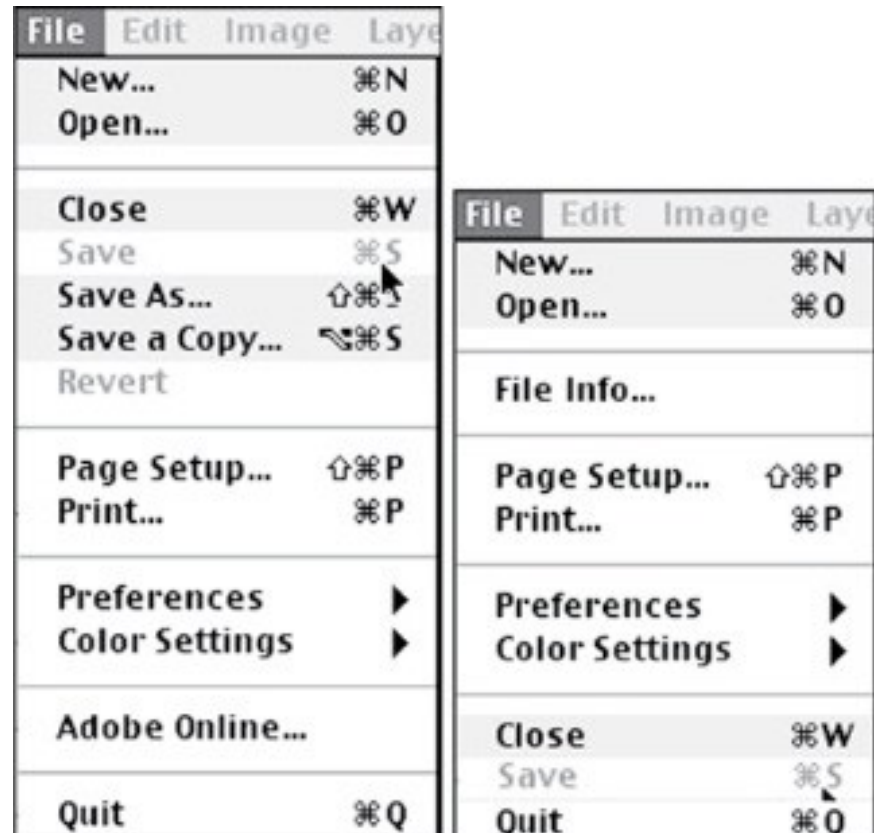
- **Utility** refers to the extent to which the product provides the right kind of functionality so that users can do what they need or want to do.
- **Question:**
 - Does the product provide an appropriate set of functions that will enable them to carry out all of their tasks in the way they want to do them?

Usability goals: Learnability

- **Learnability** refers to how easy a product is to learn to use.
- **Question:**
 - Is it possible for someone to work out basic use of the product by exploring the interface and trying certain actions?
 - How hard will it be to master the product in this way?
 - Are additional learning tools needed?

Usability goals: Learnability

Figure 1.8 (a) A safe and unsafe menu. Which is which and why? (b) A warning dialog box on macOS



Usability goals: Memorability

- **Memorability** refers to how easy a product is to remember how to use, once learned.
- **Question:**
 - What types of interface support have been provided to help someone remember how to carry out tasks, especially for ones they use infrequently?

Usability goals: satisfaction

- Satisfaction generally refers to how acceptable a product is when being used.
- It is most often used to measure a customer's experience.
- Various satisfaction scales have been developed for this purpose, for example, asking customers to give **a score from 1–5** to indicate how satisfied they are after using a product.
- **Question:**
 - What are the mean, median, and mode values on the **CSAT (Customer Satisfaction Score)** scale?

Usability and user experience goals

- Selecting terms to convey a person's feelings, emotions, and so forth can help designers understand the multifaceted nature of the user experience
- How do usability goals differ from user experience goals?
- Are there trade-offs between the two kinds of goals? (for example, can a product be both fun and safe?)
- How easy is it to measure usability versus user experience goals?

User experience goals

Desirable aspects

Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhancing sociability	Rewarding
Exciting	Supporting creativity	Emotionally fulfilling
Entertaining	Cognitively stimulating	Experiencing flow

Undesirable aspects

Boring	Unpleasant	Creepy
Frustrating	Patronizing	Intrusive
Making one feel guilty	Makes one feel stupid	Deceptive
Annoying	Cutesy	
Childish	Gimmicky	

Design principles

- Generalizable abstractions for thinking about different aspects of design
- The do's and don'ts of interaction design
- What to provide and what not to provide at the interface
- Derived from a mix of theory-based knowledge, experience, and common-sense

Visibility - poor interface



www.baddesigns.com

- This is a control panel for an elevator
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button?
Still nothing. What do you need to do?
- It is not visible as to what to do!

Visibility - Improving on a poor interface



...with this elevator, you need to insert your room card in the slot by the buttons to get the elevator to work!

How would you make this action more visible?

- Make the card reader more obvious
- Provide an auditory message that says what to do (which language?)
- Provide a big label next to the card reader that flashes when someone enters
- Make relevant parts visible
- Make what has to be done obvious

What do I do if I am wearing black?

- Invisible automatic controls can make it more difficult to use
 - people have to guess where to move their hands, bodies, or feet—on, into, or in front of—to make them work.



Fig. A sign in the restrooms at the Cincinnati airport

Feedback

- Sending information back to the user about what has been done
- Includes sound, highlighting, animation, and combinations of these
- For example, when screen button is clicked, it provides sound or red highlight feedback:

 → “ccclichhk”

 → 

Constraints

- Restricting the possible actions that can be performed
- Helps prevent user from selecting incorrect options
- Physical objects can be designed to constrain things. (for example, there being only one way you can insert a key into a lock)

Constraints

The screenshot shows the Microsoft PowerPoint interface. The 'Arrange' menu is open, displaying options for reordering and grouping objects. The slide in the background is titled 'UCLIC' and contains two bullet points. The slide has a red header bar with the 'UCL' logo and a red footer bar with the 'UCLIC' logo. The slide content is as follows:

- What was the last expensive item you bought (e.g. smartphone, bike, coat)
- How many different emotions did you go through?

The 'Arrange' menu options are:

- Reorder Objects
- Reorder Overlapping Objects
- Bring to Front
- Send to Back
- Bring Forward
- Send Backward
- Group (⌘ G)
- Ungroup (⌘ Shift G)
- Regroup (⌘ J)
- Rotate or Flip (▶)
- Align or Distribute (▶)

Logical or ambiguous design?



www.baddesigns.com

- Where do you plug the mouse?
- Where do you plug the keyboard, in the top or bottom connector?
- Do the color-coded icons help?

How to design them more logically



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(A) provides direct adjacent mapping between icon and connector



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(B) provides color coding that associates the connectors with the labels

Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks. (for example, always use Ctrl key plus first initial of the command for an operation: Ctrl+c, Ctrl+s, Ctrl+o)
- The main benefit is that consistent interfaces are easier to learn and use

When consistency breaks down

- What happens if there is more than one command starting with the same letter? (for example, save, spelling, select, style)
- You have to find other initials or combinations of keys, thereby breaking the consistency rule (for example, Ctrl+s, Ctrl+Sp, Ctrl+shift+l)
- Increases learning burden on user, making them more prone to errors

Internal and external consistency

- Internal consistency refers to designing operations to behave the same within an application
 - Difficult to achieve with complex interfaces
- External consistency refers to designing operations, interfaces, and so on to be the same across applications and devices
 - Very rarely the case, based on different designer's preference

Keypad numbers layout

A case of external inconsistency

(a) phones, remote controls

1	2	3
4	5	6
7	8	9
	0	

(b) calculators, computer keypads

7	8	9
4	5	6
1	2	3
0		

Affordances: to give a clue

- Refers to an attribute of an object that allows people to know how to use it. (For example, a mouse button invites pushing, a door handle affords pulling)
- Norman (1988) used the term to discuss the design of everyday objects
- Has since been popularized in interaction design to discuss how to design interface objects (for example, scrollbars to enable moving up and down; icons to click on)

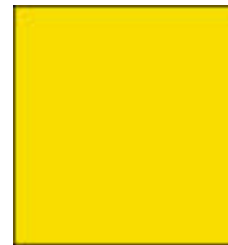
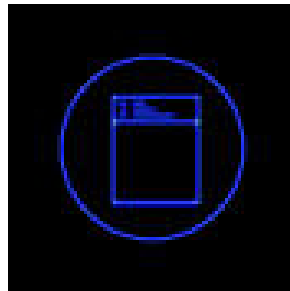
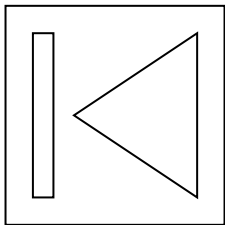
What does “affordance” have to offer interaction design?

- Interfaces are virtual and do not have affordances like physical objects
- Norman argues that it does not make sense to talk about interfaces in terms of ‘real’ affordances
- Instead, interfaces are better conceptualized as ‘perceived’ affordances:
 - Learned conventions of arbitrary mappings between action and effect at the interface
 - Some mappings are better than others

Activity

Virtual affordances

- How do these screen objects afford?
- What if you were a novice user?
- Would you know what to do with them?



Key points

- Interaction design is concerned with designing interactive products to support how people communicate and interact in their everyday and working lives
- It is concerned with how to create quality user experiences for services, devices, and interactive products
- It is multidisciplinary, involving many inputs from wide-reaching disciplines and fields
- Optimizing the interaction between users and interactive products requires consideration of a number of interdependent factors, including context of use, types of activity, UX goals, accessibility, cultural differences, and user groups.
- Design principles, such as feedback and simplicity, are useful heuristics for informing, analyzing, and evaluating aspects of an interactive product.